# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** When the Moon is not visible, it is the: 1

**(a)** full moon **(b)** new moon **(c)** gibbous moon **(d)** half moon

**2.** A lunar year of 12 months has about: 1

**(a)** 365 days **(b)** 354 days **(c)** 360 days **(d)** 400 days

**3.** The extra day in a leap year is added to the month of: 1

**(a)** December **(b)** February **(c)** March **(d)** January

**4.** The phases of the Moon are caused by: 1

**(a)** the Moon's rotation only
**(b)** clouds
**(c)** the changing positions of the Moon, Earth and Sun
**(d)** shadow of the Earth always

**5.** A calendar which combines Sun and Moon is: 1

**(a)** lunisolar **(b)** solar **(c)** lunar **(d)** digital

**6.** A simple device made of a vertical stick that shows time by its shadow is a: 1

**(a)** pendulum clock **(b)** stopwatch **(c)** sundial **(d)** hourglass

**7.** The Moon shines because it: 1

**(a)** burns **(b)** reflects sunlight
**(c)** is hot **(d)** produces its own light

**8.** The seasons occur because of the: 1

**(a)** rotation only
**(b)** tilt of the Earth's axis and its revolution
**(c)** distance from the Sun only
**(d)** Moon's pull

**9.** The shadow of a vertical stick is shortest at: 1

**(a)** noon **(b)** sunset **(c)** sunrise **(d)** midnight

**10.** The Moon shows the same face to the Earth because: 1

**(a)** it is far
**(b)** it does not rotate
**(c)** it is round
**(d)** its rotation period equals its orbital period

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Differentiate between rotation and revolution of the Earth. 2

**12.** What is the Pole Star? How does it help in direction finding? 2

**13.** What is a leap year? Why is it needed? 2

**14.** What are the phases of the Moon? 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Explain why a leap year occurs every four years. 3

**16.** State three observations that show that the stars appear to move in the night sky. 3

**17.** Why do the Hindu festivals fall on different dates in the Gregorian calendar each year? 3

OR

Describe the phases of the Moon in a lunar month with four main stages. 3

**18.** How can the Pole Star be found using the Big Dipper (Saptarishi)? 3

OR

Explain why the full moon and the new moon occur about 15 days apart. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Explain how to find direction at night using the Big Dipper and the Pole Star. Why does the Pole Star appear fixed? 5

OR

Explain the evidence that shows the Earth rotates, such as sunrise, sunset, star trails and the time differences in places. 5

**20.** Describe solar, lunar and lunisolar calendars with examples and the need for leap years or extra months. 5

OR

Discuss how the Moon's phases and the solar year were used in India to make festivals and calendars with examples of Diwali or Holi. 5

## END OF QUESTION PAPER Total: 40 Marks